

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: Multi-element ICP Standard, Specpure
Cat No. : 98119

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company Avocado Research Chemicals Ltd.
(Part of Thermo Fisher Scientific)
Shore Road, Heysham
Lancashire, LA3 2XY,
United Kingdom
Office Tel: +44 (0) 1524 850506
Office Fax: +44 (0) 1524 850608

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Physical hazards

Substances/mixtures corrosive to metal Category 1 (H290)

Health hazards

Acute Inhalation Toxicity - Vapors Category 4 (H332)
Skin Corrosion/Irritation Category 1 (H314) B
Serious Eye Damage/Eye Irritation Category 1 (H318)

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Carcinogenicity	Category 1B (H350)
<u>Environmental hazards</u>	
Chronic aquatic toxicity	Category 3 (H412)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals
H332 - Harmful if inhaled
H314 - Causes severe skin burns and eye damage
H350 - May cause cancer
H412 - Harmful to aquatic life with long lasting effects
EUH208 - Contains Chromium. May produce an allergic reaction

Precautionary Statements

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

Additional EU labelling

Restricted to professional users

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567
Water	7732-18-5	231-791-2	93.4	-
Nitric acid ...% [C ≤ 70 %]	7697-37-2	231-714-2	5	Ox. Liq. 3 (H272) Met. Corr. 1 (H290) Acute Tox. 3 (H331) Skin Corr. 1A (H314) Eye Dam. 1 (H318) (EUH071)
Selenium	7782-49-2	EEC No. 231-957-4	0.5	Acute Tox. 3 (H331) Acute Tox. 3 (H301)

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				STOT RE 2 (H373) Aquatic Chronic 4 (H413)
Arsenic	7440-38-2	EEC No. 231-148-6	0.5	Acute Tox. 3 (H301) Acute Tox. 3 (H331) Carc. 1A (H350) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Silver	7440-22-4	EEC No. 231-131-3	0.1	-
Nickel	7440-02-0	EEC No. 231-111-4	0.1	Skin Sens. 1 (H317) Carc. 2 (H351) STOT RE 1 (H372)
Lead	7439-92-1	EEC No. 231-100-4	0.1	Acute Tox. 4 (H332) Acute Tox. 4 (H302) Repr. 1A (H360Df) STOT RE 2 (H373) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Chromium	7440-47-3	EEC No. 231-157-5	0.1	-
Cadmium	7440-43-9	EEC No. 231-152-8	0.1	Acute Tox. 2 (H330) Muta. 2 (H341) Carc. 1B (H350) Repr. 2 (H361fd) STOT RE 1 (H372) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Barium	7440-39-3	EEC No. 231-149-1	0.1	Flam. Sol. 2 (H228) Water-react. 1 (H260) Acute Tox. 3 (H301) Skin Corr. 1B (H314) Eye Dam. 1 (H318)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Nitric acid ...% [C ≤ 70 %]	Ox. Liq. 2 :: C ≥ 99% Ox. Liq. 3 :: 65% ≤ C < 99% Acute Tox. 1 (inhal) :: C ≥ 70% Acute Tox. 3 (inhal) :: 70% > C ≥ 26.5% Acute Tox. 4 (inhal) :: 26.5% > C ≥ 13.25% Skin Corr. 1A :: C ≥ 20% Skin Corr. 1B :: 5% ≤ C < 20% Met. Corr. 1 :: C ≥ 2% EUH071 :: C ≥ 20%	-	-
Arsenic	-	1	-
Lead	Repr. 1A : C ≥ 0.03 % STOT RE 1 : C ≥ 0.5 %	1 (acute) 10 (Chronic)	-
Cadmium	-	10	-

Component	ECHA (RAC) ATE (Oral)	ECHA (RAC) ATE (Dermal)	ECHA (RAC) ATE (Inhalation)
Nitric acid ...% [C ≤ 70 %]	-	-	ATE = 2.65 mg/L (vapours)

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Remove and wash

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	contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.
Ingestion	Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.
Inhalation	If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

Hazardous Combustion Products

Nitrogen oxides (NO_x), Metal oxides.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

6.3. Methods and material for containment and cleaning up

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Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Class 6.1D
Storage Class (LGK) (Germany)

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

Component	The United Kingdom	European Union	Ireland
Nitric acid ...% [C ≤ 70 %]	STEL: 1 ppm 15 min STEL: 2.6 mg/m ³ 15 min	STEL: 1 ppm (15min) STEL: 2.6 mg/m ³ (15min)	STEL: 1 ppm 15 min STEL: 2.6 mg/m ³ 15 min
Selenium	STEL: 0.3 mg/m ³ 15 min TWA: 0.1 mg/m ³ 8 hr		TWA: 0.1 mg/m ³ 8 hr. Se STEL: 0.3 mg/m ³ 15 min
Arsenic	STEL: 0.3 mg/m ³ 15 min TWA: 0.1 mg/m ³ 8 hr Carc.		TWA: 0.01 mg/m ³ 8 hr. STEL: 0.03 mg/m ³ 15 min
Silver	STEL: 0.3 mg/m ³ 15 min TWA: 0.1 mg/m ³ 8 hr	TWA: 0.1 mg/m ³ (8h)	TWA: 0.1 mg/m ³ 8 hr. Ag metallic STEL: 0.3 mg/m ³ 15 min
Nickel	STEL: 1.5 mg/m ³ 15 min TWA: 0.5 mg/m ³ 8 hr Skin		TWA: 0.5 mg/m ³ 8 hr. STEL: 1.5 mg/m ³ 15 min
Lead	STEL: 0.45 mg/m ³ 15 min TWA: 0.15 mg/m ³ 8 hr	TWA: 0.15 mg/m ³ (8h)	TWA: 0.15 mg/m ³ 8 hr. STEL: 0.45 mg/m ³ 15 min
Chromium	STEL: 1.5 mg/m ³ 15 min TWA: 0.5 mg/m ³ 8 hr	TWA: 2 mg/m ³ (8hr)	TWA: 2 mg/m ³ 8 hr. STEL: 6 mg/m ³ 15 min
Cadmium	STEL: 0.075 mg/m ³ 15 min TWA: 0.025 mg/m ³ 8 hr Carc. metal	TWA: 0.001 mg/m ³ (8h)	TWA: 0.001 mg/m ³ 8 hr. inhalable fraction TWA: 0.004 mg/m ³ 8 hr. limit

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			value 0.004 mg/m ³ until 11 July 2027 inhalable fraction STEL: 0.003 mg/m ³ 15 min STEL: 0.012 mg/m ³ 15 min
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Biological limit values

List source(s):

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Selenium 7782-49-2 (0.5)				DNEL = 7mg/kg bw/day
Arsenic 7440-38-2 (0.5)				DNEL = 85µg/kg bw/day
Nickel 7440-02-0 (0.1)			DNEL = 0.035mg/cm2	
Barium 7440-39-3 (0.1)				DNEL = 28.5mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Selenium 7782-49-2 (0.5)				DNEL = 0.05mg/m ³
Arsenic 7440-38-2 (0.5)				DNEL = 4µg/m ³
Silver 7440-22-4 (0.1)				DNEL = 0.1mg/m ³
Nickel 7440-02-0 (0.1)	DNEL = 11.9mg/m ³		DNEL = 0.05mg/m ³	DNEL = 0.05mg/m ³
Chromium 7440-47-3 (0.1)			DNEL = 0.5mg/m ³	
Cadmium 7440-43-9 (0.1)			DNEL = 4µg/m ³	
Barium 7440-39-3 (0.1)				DNEL = 5.8mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Arsenic 7440-38-2 (0.5)	PNEC = 13µg/L	PNEC = 130mg/kg sediment dw	PNEC = 13µg/L	PNEC = 61µg/L	PNEC = 0.5mg/kg soil dw
Silver 7440-22-4 (0.1)	PNEC = 0.04µg/L	PNEC = 438.13mg/kg sediment dw		PNEC = 0.025mg/L	PNEC = 1.41mg/kg soil dw
Nickel 7440-02-0 (0.1)	PNEC = 7.1µg/L	PNEC = 109mg/kg sediment dw		PNEC = 0.33mg/L	PNEC = 29.9mg/kg soil dw
Lead 7439-92-1 (0.1)	PNEC = 2.4µg/L	PNEC = 186mg/kg sediment dw		PNEC = 100µg/L	PNEC = 212mg/kg soil dw
Chromium 7440-47-3 (0.1)	PNEC = 6.5µg/L	PNEC = 205.7mg/kg sediment dw			PNEC = 21.1mg/kg soil dw
Cadmium 7440-43-9 (0.1)	PNEC = 0.19µg/L	PNEC = 1.8mg/kg sediment dw		PNEC = 20µg/L	PNEC = 0.9mg/kg soil dw
Barium 7440-39-3 (0.1)	PNEC = 114.7µg/L	PNEC = 598.9mg/kg		PNEC = 62.2mg/L	PNEC = 207.7mg/kg soil dw

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		sediment dw			
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Component	Marine water	Marine water sediment	Marine water intermittent	Food chain	Air
Arsenic 7440-38-2 (0.5)	PNEC = 0.8µg/L	PNEC = 8mg/kg sediment dw	PNEC = 0.9µg/L	PNEC = 1mg/kg food	
Silver 7440-22-4 (0.1)	PNEC = 0.86µg/L	PNEC = 438.13mg/kg sediment dw			
Nickel 7440-02-0 (0.1)	PNEC = 8.6µg/L	PNEC = 109mg/kg sediment dw		PNEC = 0.12mg/kg food	
Lead 7439-92-1 (0.1)	PNEC = 3.3µg/L	PNEC = 168mg/kg sediment dw		PNEC = 10.9mg/kg food	
Cadmium 7440-43-9 (0.1)	PNEC = 1.14µg/L	PNEC = 0.64mg/kg sediment dw		PNEC = 0.16mg/kg food	

8.2. Exposure controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber	See manufacturers	-	EN 374	(minimum requirement)
Nitrile rubber	recommendations			
Neoprene				
PVC				

Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

Recommended Filter type: Particulates filter conforming to EN 143

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Recommended half mask:- Particle filtering: EN149:2001

When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

Prevent product from entering drains.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance		
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	2	
Viscosity	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Nitric acid ...% [C ≤ 70 %]	-2.3	
Vapor Pressure	No data available	
Density / Specific Gravity	No data available	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization

No information available.

Hazardous Reactions

None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Metal oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

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Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Nitric acid ...% [C ≤ 70 %]	-	-	LC50 = 2500 ppm. (Rat) 1h
Selenium	LD50 = 6700 mg/kg (Rat)	-	>5.67 mg/l (Rat) 4hr
Arsenic	LD50 = 15 mg/kg (Rat)	-	-
Silver	> 2000 mg/kg (Rat)	LD50 > 2000 mg/kg (rat)	LC50 > 5.16 mg/L (Rat) 4 h
Nickel	LD50 > 9000 mg/kg (Rat)	-	LC50 > 10.2 mg/L (Rat) 1 h
Cadmium	LD50 = 2330 mg/kg (Rat)	-	LC50 = 25 mg/m ³ (Rat) 30 min
Barium	LD50 = 132 mg/kg (Rat)	-	-

Component	ECHA (RAC) ATE (Oral)	ECHA (RAC) ATE (Dermal)	ECHA (RAC) ATE (Inhalation)
Nitric acid ...% [C ≤ 70 %]	-	-	ATE = 2.65 mg/L (vapours)

(b) skin corrosion/irritation;

Category 1 B

(c) serious eye damage/irritation;

Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity;

No data available

(f) carcinogenicity;

Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Arsenic			Cat. 1	Group 1
Nickel			Cat. 1	Group 2B
Lead				Group 2A
Cadmium	Carc Cat. 1B		Cat. 1	Group 1

(g) reproductive toxicity;

No data available

(h) STOT-single exposure;

No data available

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

No data available

Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

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11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Selenium	LC50: > 100 mg/L, 96h semi-static (Oncorhynchus mykiss)		
Silver	LC50: = 0.064 mg/L, 96h static (Lepomis macrochirus) LC50: = 0.0062 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: 0.00155 - 0.00293 mg/L, 96h static (Pimephales promelas)	EC50: = 0.00024 mg/L, 48h Static (Daphnia magna)	
Nickel	LC50: > 100 mg/L, 96h (Brachydanio rerio) LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio) LC50: = 10.4 mg/L, 96h static (Cyprinus carpio)	EC50 = 510 µg/L 96h	EC50 = 0.1 mg/L 72h EC50 = 0.18 mg/L 72h
Lead	LC50: = 1.32 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 1.17 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: = 0.44 mg/L, 96h semi-static (Cyprinus carpio)	EC50: = 600 µg/L, 48h (water flea)	
Cadmium	LC50: 0.0004 - 0.003 mg/L, 96h (Pimephales promelas) LC50: = 0.016 mg/L, 96h (Oryzias latipes) LC50: = 21.1 mg/L, 96h flow-through (Lepomis macrochirus) LC50: = 0.24 mg/L, 96h static (Cyprinus carpio) LC50: = 4.26 mg/L, 96h semi-static (Cyprinus carpio) LC50: = 0.002 mg/L, 96h (Cyprinus carpio) LC50: = 0.006 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 0.003 mg/L, 96h flow-through (Oncorhynchus mykiss)	EC50: = 0.0244 mg/L, 48h Static (Daphnia magna)	
Barium	LC50: > 500 mg/L/96h (Cyprinodon variegatus)		

Component	Microtox	M-Factor
Arsenic		1

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Lead		1 (acute) 10 (Chronic)
Cadmium		10

12.2. Persistence and degradability	Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary
Persistence	May persist.
Degradation in sewage treatment plant	Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential	Product has a high potential to bioconcentrate
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Component	log Pow	Bioconcentration factor (BCF)
Nitric acid ...% [C ≤ 70 %]	-2.3	No data available
Chromium		1.03 - 1.22

12.4. Mobility in soil	No information available
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12.5. Results of PBT and vPvB assessment	No data available for assessment.
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12.6. Endocrine disrupting properties	
Endocrine Disruptor Information	This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects	
Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN3264
14.2. UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s.
Technical Shipping Name	Nitric acid solution
14.3. Transport hazard class(es)	8
14.4. Packing group	III

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ADR

14.1. UN number	UN3264
14.2. UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s.
Technical Shipping Name	Nitric acid solution
14.3. Transport hazard class(es)	8
14.4. Packing group	III

IATA

14.1. UN number	UN3264
14.2. UN proper shipping name	Corrosive liquid, acidic, inorganic, n.o.s.
Technical Shipping Name	Nitric acid solution
14.3. Transport hazard class(es)	8
14.4. Packing group	III

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-
Nitric acid ...% [C ≤ 70 %]	7697-37-2	231-714-2	-	-	X	X	KE-25911	X	X
Selenium	7782-49-2	231-957-4	-	-	X	X	KE-30924	X	-
Arsenic	7440-38-2	231-148-6	-	-	X	X	KE-01933	X	-
Silver	7440-22-4	231-131-3	-	-	X	X	KE-31261	X	-
Nickel	7440-02-0	231-111-4	-	-	X	X	KE-25818	X	-
Lead	7439-92-1	231-100-4	-	-	X	X	KE-21887	X	-
Chromium	7440-47-3	231-157-5	-	-	X	X	KE-05970	X	-
Cadmium	7440-43-9	231-152-8	-	-	X	X	KE-04397	X	-
Barium	7440-39-3	231-149-1	-	-	X	X	KE-02022	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Water	7732-18-5	X	ACTIVE	X	-	X	X	X
Nitric acid ...% [C ≤ 70 %]	7697-37-2	X	ACTIVE	X	-	X	X	X
Selenium	7782-49-2	X	ACTIVE	X	-	X	X	X
Arsenic	7440-38-2	X	ACTIVE	X	-	X	X	X
Silver	7440-22-4	X	ACTIVE	X	-	X	X	X
Nickel	7440-02-0	X	ACTIVE	X	-	X	X	X
Lead	7439-92-1	X	ACTIVE	X	-	X	X	X
Chromium	7440-47-3	X	ACTIVE	X	-	X	X	X
Cadmium	7440-43-9	X	ACTIVE	X	-	X	X	X
Barium	7440-39-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

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Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Nitric acid ...% [C ≤ 70 %]	7697-37-2	-	Use restricted. See item 75. (see link for restriction details)	-
Selenium	7782-49-2	-	Use restricted. See item 75. (see link for restriction details)	-
Arsenic	7440-38-2	-	Use restricted. See item 75. (see link for restriction details)	-
Silver	7440-22-4	-	Use restricted. See item 75. (see link for restriction details)	-
Nickel	7440-02-0	-	Use restricted. See item 27. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	-
Lead	7439-92-1	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 30. (see link for restriction details) Use restricted. See item 63. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 231-100-4 - Toxic for reproduction (Article 57c)
Chromium	7440-47-3	-	Use restricted. See item 75. (see link for restriction details)	-
Cadmium	7440-43-9	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 23. (see link for restriction details) Use restricted. See item 28. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 231-152-8 - Carcinogenic, Article 57a; Specific target organ toxicity after repeated exposure, Article 57(f) - human health
Barium	7440-39-3	-	Use restricted. See item 75. (see link for restriction details)	-

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in

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scientific research and development which includes routine analytics or use as intermediate.

REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Water	7732-18-5	Not applicable	Not applicable
Nitric acid ...% [C ≤ 70 %]	7697-37-2	Not applicable	Not applicable
Selenium	7782-49-2	Not applicable	Not applicable
Arsenic	7440-38-2	Not applicable	Not applicable
Silver	7440-22-4	Not applicable	Not applicable
Nickel	7440-02-0	Not applicable	Not applicable
Lead	7439-92-1	Not applicable	Not applicable
Chromium	7440-47-3	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	Not applicable
Barium	7440-39-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Component	ANNEX I - PART 1 List of chemicals subject to export notification procedure (referred to in Article 8)	ANNEX I - PART 2 List of chemicals qualifying for PIC notification (referred to in Article 11)	ANNEX I - PART 3 List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14)
Lead 7439-92-1 (0.1)	sr — severe restriction i(2) — industrial chemical for public	-	-
Cadmium 7440-43-9 (0.1)	i(1) — industrial chemical for professional use sr — severe restriction i(2) — industrial chemical for public sr — severe restriction	i — industrial chemical sr — severe restriction	-

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Nitric acid ...% [C ≤ 70 %]	WGK1	
Selenium	WGK2	Class II : 0.5 mg/m³ (Massenkonzentration)
Arsenic	WGK3	Krebserzeugende Stoffe - Class I : 0.05 mg/m³ (Massenkonzentration)
Silver	nwg	

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Nickel	WGK 2	Class II : 0.5 mg/m ³ (Massenkonzentration) Krebserzeugende Stoffe - Class II : 0.5 mg/m ³ (Massenkonzentration)
Lead	nwg	Class II : 0.5 mg/m ³ (Massenkonzentration)
Chromium	nwg	Class III : 1 mg/m ³ (Massenkonzentration)
Cadmium	WGK3	Krebserzeugende Stoffe - Class I : 0.05 mg/m ³ (Massenkonzentration)
Barium	WGK1	

Component	France - INRS (Tables of occupational diseases)
Selenium	Tableaux des maladies professionnelles (TMP) - RG 75
Arsenic	Tableaux des maladies professionnelles (TMP) - RG 20,RG 20bis
Lead	Tableaux des maladies professionnelles (TMP) - RG 1
Chromium	Tableaux des maladies professionnelles (TMP) - RG 10
Cadmium	Tableaux des maladies professionnelles (TMP) - RG 61,RG 61bis

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Nitric acid ...% [C ≤ 70 %] 7697-37-2 (5)	Prohibited and Restricted Substances		
Arsenic 7440-38-2 (0.5)	Prohibited and Restricted Substances		Annex I - pesticide
Nickel 7440-02-0 (0.1)	Prohibited and Restricted Substances		
Lead 7439-92-1 (0.1)	Prohibited and Restricted Substances		
Chromium 7440-47-3 (0.1)	Prohibited and Restricted Substances		
Cadmium 7440-43-9 (0.1)	Prohibited and Restricted Substances		Annex I - industrial chemical

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals
 H332 - Harmful if inhaled
 H314 - Causes severe skin burns and eye damage
 H318 - Causes serious eye damage
 H350 - May cause cancer
 H412 - Harmful to aquatic life with long lasting effects
 H260 - In contact with water releases flammable gases which may ignite spontaneously
 H272 - May intensify fire; oxidizer
 H301 - Toxic if swallowed
 H302 - Harmful if swallowed
 H317 - May cause an allergic skin reaction
 H330 - Fatal if inhaled
 H331 - Toxic if inhaled
 H341 - Suspected of causing genetic defects
 H351 - Suspected of causing cancer
 H360Df - May damage the unborn child. Suspected of damaging fertility
 H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child
 H372 - Causes damage to organs through prolonged or repeated exposure
 H400 - Very toxic to aquatic life
 H410 - Very toxic to aquatic life with long lasting effects

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EUH071 - Corrosive to the respiratory tract

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Prepared By Health, Safety and Environmental Department

Revision Date 15-Feb-2024

Revision Summary New emergency telephone response service provider.

This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet